Exam 1 - Requires Respondus LockDown Browser

- Due Oct 14 at 11:59pm
- Points 22
- Questions 22
- Available Oct 12 at 12am - Oct 14 at 11:59pm
- Time Limit 75 Minutes
- Requires Respondus LockDown Browser

Instructions

Exam duration is 75-minutes maximum.

Students are allowed

- Scientific/graphing/financial calculator
- One 8.5"x11" cheat sheet with writing/typing on both sides
- Scratch paper

Scratch paper and cheat sheet will be collected at the conclusion of the exam.

Scratch papers will not be graded.

All questions are worth 1 point each. There are 22 questions. Your final grade is min(20, raw grade).

This quiz was locked Oct 14 at 11:59pm.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	75 minutes	15 out of 22

Score for this quiz: 15 out of 22
Submitted Oct 14 at 6:35pm
This attempt took 75 minutes.

For numerical answers, round to three decimal places if the exact answer has more than three decimal places.
Wrong answer

Question 1
0 / 1 pts

A company is organizing a small project team consisting of five distinct roles across three specialties.

- Two people must be selected from a group of four analysts to fill the roles of **Lead Analyst** and **Assistant Analyst.**
- Two people must be selected from a group of three engineers to fill the roles of **Senior Engineer** and **Junior Engineer**.
- One person must be selected from a group of five designers to serve as **Graphic Designer**.

Assume that the three groups do not overlap and each person can hold at most one role. Which of the following is the number of different ways the team can be formed?

- 90
- 180
- 720
- 360
- 23

Because each of the first two specialties has **two distinct roles**, we must count **assignments** of people to roles (order matters), i.e., permutations.

Analysts: $P(4,2) = 12$

Engineers: $P(3,2) = 6$

Designers: $P(5,1) = 5$

Multiply the independent choices to obtain the possible ways to form a team. $12 \times 6 \times 5 = 360$

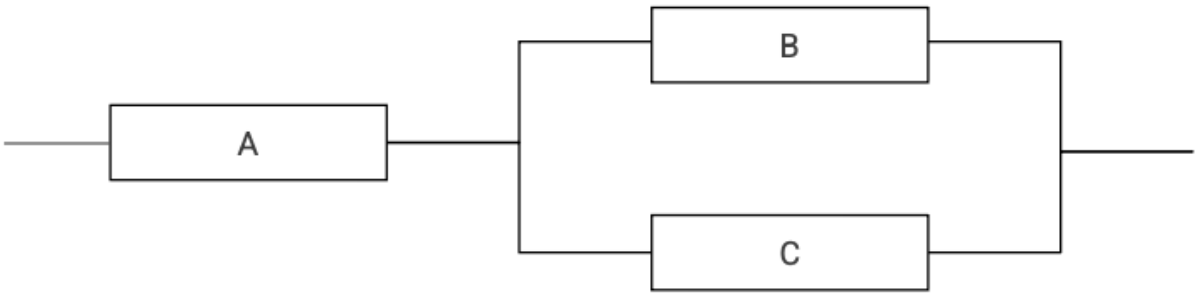
Wrong answer

Question 2
0 / 1 pts

The functioning states of components B and C are **dependent**, with joint probabilities given in the table below. Component A operates **independently** of B and C and has reliability 0.8.

Probability	B functions	B fails
C functions	0.4	0.2
C fails	0.3	0.1

Component A is connected in series with a parallel arrangement of B and C, as shown in the diagram below. Which of the following is the reliability of the system?



- 0.72
- 0.32
- 0.90
- 0.704
- 0.336

The system works if and only if A works **AND** at least one of B and C functions. That is, the reliability of the system is $\mathbb{P}($ A functions and at least one of B and C functions$)$. Since A operates independently of both B and C, we can further write

$\mathbb{P}($A functions and at least one of B and C functions$)= \mathbb{P}($A functions$)\times\mathbb{P}($at least one of B and C functions$)$.

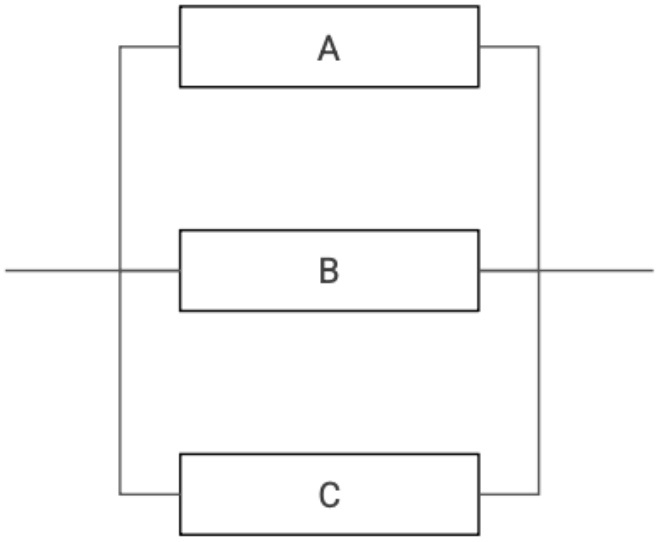
$\mathbb{P}($A functions$)= 0.8$. $\mathbb{P}($at least one of B and C functions$)= 1-\mathbb{P}($both B and C fail$)= 1-0.1=0.9$.

Thus the reliability is 0.9×0.8

Correct answer

Question 3
1 / 1 pts

In contrast to the previous problem, assume that components A, B, and C are **independent**, with reliabilities 0.9, 0.6 and 0.5, respectively. Determine the reliability of new system below.



- 0.27
- 0.73
- 0.02
- 0.98
- 0.95

$1-\mathbb{P}(\text{All three fail})=1-(1-0.9)\times(1-0.6)\times(1-0.5)=0.98$

Wrong answer

Question 4
0 / 1 pts

At a particular University, 2/3 of computer science students are familiar with Python for programming, and 1/4 are familiar with C++. What is the **largest** possible fraction of students who are familiar with **neither**?

- 11/12
- 1/3
- 1/12
- 3/4
- 1/6

When the 1/4 of students familiar with C++ are also familiar with Python, 2/3 of students are familiar with both and the rest 1/3 are familiar with neither.

Wrong answer

Question 5
0 / 1 pts

In the previous problem, it turns out that, in fact, 1/4 of students are familiar with **neither**. Are the two events (familiar with Python and familiar with C++) independent?

- No, they are NOT independent
- Not enough information to determine if they are independent.
- Yes, they are independent

$1-\frac{1}{4}-\frac{2}{3}-\frac{1}{4}+\mathbb{P}(\textbf{Both})=0\Rightarrow\mathbb{P}(\textbf{Python and C++})=\frac{1}{6}.$

$\mathbb{P}(\textbf{Python})\times\mathbb{P}(\textbf{C++})=\frac{2}{3}\times\frac{1}{4}=\frac{1}{6}=\mathbb{P}(\textbf{Python and C++})\Rightarrow$Independent

Wrong answer

Question 6
0 / 1 pts

A newly launched daily flight route is being monitored by an airline for operational reliability. Historical data suggest that each flight is delayed independently with probability 0.3.

Consider the following two probabilities p_1 and p_2 where

- p_1: the first delay occurs on the **third day** of operation
- p_2: given that there is no delay for the first two days, the probability that a delay occurs on the next day.

Which probability is greater?

- p_1
- $p_1=p_2$
- p_2

$p_1=(1-0.3)\times(1-0.3)\times0.3=0.147.$

$p_2=\mathbb{P}(\textbf{3rd delay}|\textbf{1st no delay, 2nd no delay})=\mathbb{P}(\textbf{3rd day delay})=0.3$ by the independence of flight operation.

Correct answer

Question 7
1 / 1 pts

A flight has 250 seats, and all seats have been sold in advance. Historically, each passenger independently shows up for departure with a fixed probability $0<p<1$ (the show-up rate). Let X denote the number of ticketed passengers who show up. What is an appropriate probability distribution for X?

- Poisson distribution

- Geometry distribution
- Binomial distribution
- Bernoulli

$X \sim B(250, p)$

Correct answer

Question 8
1 / 1 pts

In the previous problem, if the expected value $\mathbb{E}(X) = 200$, what is the variance $Var(X)$?

40

40 (with margin: 1)

Since $X \sim B(250, p)$, $\mathbb{E}[X] = 200$ implies $p = 0.8$. Then $Var(X) = 250 \times p \times (1 - p) = 40$

Wrong answer

Question 9
0 / 1 pts

A fair six-sided die is rolled by three players sequentially. Let X denote the number of times a number greater than 4 is rolled. What is the sample space of this experiment?

- $\{(i, j, k) : 1 \leq i \leq 6, 1 \leq j \leq 6, 1 \leq k \leq 6\}$
- $\{0, 1, 2, 3\}$
- $\{1, 2, 3, \ldots 216\}$
- $\{1, 2, 3, 4, 5, 6\}$

Correct answer

Question 10
1 / 1 pts

Find $\mathbb{P}(X = 2)$ in the previous problem.

- 0.333
- 0.037
- 0.111
- 0.222
- 0.444

Exactly two of the three rolls must be greater than 4 (probability 1/3 each) and one must be 4 or less (probability 2/3); the smaller roll can occur in any of the three positions.

Thus, $\mathbb{P}(X = 2) = 3 \times \left(\frac{1}{3}\right)^2 \times \frac{2}{3} = \frac{6}{27} = 0.222$

Correct answer

Question 11
1 / 1 pts

Continuing from the previous problem, let Y_1, Y_2, and Y_3 denote the outcomes of the first, second, and third players, respectively. What is the probability that (Y_1, Y_2, Y_3) consists of **three consecutive integers in increasing order**, i.e., $Y_2 = Y_1 + 1$ and $Y_3 = Y_2 + 1$. For example, $(Y_1, Y_2, Y_3) = (4, 5, 6)$ qualifies.

- 0.20
- 0.111
- 0.019
- 0.028

There are 6^3 possible outcomes and 4 qualifying sequences: $(1,2,3), (2,3,4), (3,4,5), (4,5,6)$.

Correct answer

Question 12

1 / 1 pts

Let X and Y be two random variables whose joint probabilities are given below. Determine the value of ??.

Probabilities	$X=-1$	$X=-2$	$X=-3$	$X=-4$
$Y=1$	0.06	0.10	0.12	0.08
$Y=2$	??	0.10	0.08	0.12
$Y=3$	0.07	0.05	0.08	0.09

0.05

0.05 (with margin: 0.01)

??=0.05 since all entries must sum to 1

Correct answer

Question 13

1 / 1 pts

Continuing from the previous problem. What is $\mathbb{P}(X+Y=0)$?

0.24

0.24 (with margin: 0.01)

$$\mathbb{P}(X+Y=0) = \mathbb{P}(X=-1, Y=1) + \mathbb{P}(X=-2, Y=2) + \mathbb{P}(X=-3, Y=3) = 0.06 + 0.10 + 0.08$$

Correct answer

Question 14

1 / 1 pts

Continuing from the previous problem, what is $\mathbb{P}(X+Y=0|X=-4)$?

0

0 (with margin: 0)

$$\mathbb{P}(X+Y=0|X=-4) = \frac{\mathbb{P}(X+Y=0, X=-4)}{\mathbb{P}(X=-4)} = \frac{\mathbb{P}(Y=4, X=-4)}{\mathbb{P}(X=-4)} = \frac{0}{0.29}$$

Correct answer

Question 15
1 / 1 pts

A hotel has 300 rooms available for a particular night, but it routinely overbooks because some guests do not show up. Assume each guest checks in independently with probability p. The hotel charges the room rate only if the guest checks in (no-shows are not charged), and ignore any costs associated with turning guests away.

Consider the following two overbooking strategies:

- **Case 1:** 320 tickets are sold at \$100 each. The check-in rate p is 0.9.
- **Case 2:** 350 tickets are sold at \$90 each. The check-in rate p is 0.85.

Compare the expected room revenue under the two cases. Which strategy yields the higher expected revenue?

- Case 1
- Same expected revenue
- Case 2

Let X_1 and X_2 denote the number of ticketed passengers that show up under the two cases, respectively. Then $X_1 \sim B(320, 0.9), X_2 \sim B(350, 0.85)$.

Case 1 revenue is $\mathbb{E}[100X_1] = 100 \times n \times p = 100 \times 320 \times 0.9 = 28800$.

Case 2 revenue is $\mathbb{E}[90X_2] = 90 \times n \times p = 90 \times 350 \times 0.85 = 26775$

Correct answer

Question 16
1 / 1 pts

A warehouse contains 12 identical-looking hard drives, 5 of which are defective. If 7 drives are randomly selected for inspection, what is the probability that exactly 3 defective drives are selected? Assume all subsets of 7 drives are equally likely to be chosen.

0.442

Between 0.43 and 0.455

$$\mathbb{P}(\text{exactly 3 defective}) = \frac{\binom{5}{3}\binom{7}{4}}{\binom{12}{7}} = 0.442.$$

Correct answer

Question 17
1 / 1 pts

Continuing from the previous problem, we instead randomly select one drive from the 12 at a time and return it to the warehouse after inspection so that it is indistinguishable from the others. Let X denote the number of defective drives observed after 7 selections. Then X follows a binomial distribution, $X \sim B(n, p)$. What are n and p?

- $n = 12, p = 0.417$
- $n = 12, p = 0.714$
- $n = 7, p = 0.417$

- $n = 7, p = 0.583$
- $n = 7, p = 0.714$
- $n = 12, p = 0.583$

Correct answer

Question 18
1 / 1 pts

In a data center, highest-priority tasks arrive as a Poisson process with parameter $\lambda = 0.3$ per hour. A monitoring system sends an alert whenever a highest-priority task arrives. During a maintenance window lasting 3 hours, what is the probability that **at least one alert** is triggered?

(Note that calculators use LN for denoting logarithm base e)

0.593

0.59 (with margin: 0.02)

$1 - \mathbb{P}(0 \text{ alert}) = 1 - e^{-(0.3 \times 3)}$

Wrong answer

Question 19
0 / 1 pts

Two dedicated servers take turns processing the highest-priority tasks received at the data center. Server 1 processes the first task, Server 2 processes the second task, then Server 1 processes the third task, and so on. Whenever a server processes a task, it fails with a fixed probability: Server 1 fails with probability 0.25, and Server 2 fails with probability 0.1. Assume failures occur independently across tasks and between servers. What is the probability that **Server 2 fails before Server 1**?

0.075

0.23 (with margin: 0.03)

Let A denote the event of failure of Server 1 and B the failure of Server 2. The cases that Server 2 fails before Server 1 could be

$A^c B$
$A^c B^c A^c B$
$A^c B^c A^c B^c A^c B$
$\vdots$

Thus the probability is
$0.75 \times 0.1 + 0.75 \times 0.9 \times 0.75 \times 0.1 + (0.75 \times 0.9)^2 \times 0.75 \times 0.1 + \cdots = \frac{1}{1-0.75\times0.9} \times 0.75 \times 0.1 = 0.231$

Correct answer

Question 20
1 / 1 pts

In the data center, an intrusion detection system (IDS) flags incoming tasks as “malicious” by raising an alert.

- If a task is truly malicious, the system raises an alert with probability 0.95 (sensitivity)

- If a task is normal, the system incorrectly raises an alert with probability 0.1 (false positive rate)
- Only 5% of all incoming tasks are actually malicious

A task triggers an alert. Let p denote the probability that the task is actually malicious. Select the interval that contains p.

- $0 < p \leq 0.2$
- $0.2 < p \leq 0.4$
- $0.8 < p \leq 1.0$
- $0.6 < p \leq 0.8$
- $0.4 < p \leq 0.6$

Let M denotes the event the incoming task is actually malicious and A denote the event the alert is triggered. Then we have the following

$$\mathbb{P}(A|M) = 0.95, \mathbb{P}(A|M^c) = 0.1, \mathbb{P}(M) = 0.05.$$

By Bayes Rule,

$$\mathbb{P}(M|A) = \frac{\mathbb{P}(A|M)\mathbb{P}(M)}{\mathbb{P}(A)} = \frac{\mathbb{P}(A|M)\mathbb{P}(M)}{\mathbb{P}(A|M)\mathbb{P}(M)+\mathbb{P}(A|M^c)\mathbb{P}(M^c)}$$
$$= \frac{0.95\times0.05}{0.95\times0.05+0.1\times(1-0.05)} = 0.33.$$

Correct answer

Question 21
1 / 1 pts

Consider the following joint probabilities for a power supply outage and a cooling system failure in the data center.

Probabilities	Power outage	**No** power outage
Cooling failure	0.12	0.28
No cooling failure	??	0.42

What is the **marginal** probability of power outage?

0.3

0.3 (with margin: 0)

?? = 0.18 since the sum of the table must be 1. Then $\mathbb{P}(\text{power outage}) = 0.12 + 0.18 = 0.3$

Correct answer

Question 22
1 / 1 pts

Based on the table in the previous problem, are power supply outage and cooling system failure independent events?

- No, they are NOT independent
- Yes, they are independent
- Not enough information to determine if they are independent.

Independence can be verified by

P(**No** power outage and Cooling failure) = 0.28

P(**No** power outage) P(Cooling failure) = (0.28 + 0.42) (0.12 + 0.28) = 0.7 * 0.4 = 0.28

Quiz Score: 15 out of 22